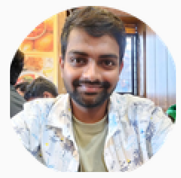
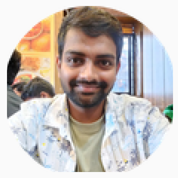


KETAN SAGARE

AI Interview Question Series

GenAI Interview Question Asked for an AI Engineer role at Amazon

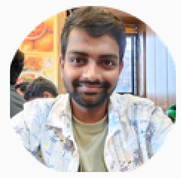


KETAN SAGARE

If you have search results from multiple retrival methods, how would you merge and homogenize the rankings into a single result set?

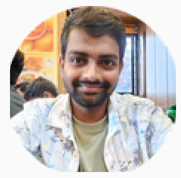
When combining search results from multiple retrieval methods in a RAG system, the goal is to create a unified ranking that preserves the strengths of each retrieval strategy. Typically, these methods may include dense vector search, keyword search like BM25, metadata filtering, or hybrid retrieval.

My first step would be score normalization because different retrieval systems produce scores on different scales. For example, cosine similarity from vector search ranges differently than BM25 relevance scores. I would normalize all scores using techniques such as Min-Max scaling or Z-score normalization so that they become comparable.



Next, I would apply rank fusion techniques. One commonly used approach is Reciprocal Rank Fusion (RRF), where documents ranked highly by multiple retrieval systems receive stronger combined scores. RRF is robust because it does not depend heavily on raw score calibration and works well across heterogeneous retrieval methods.

nother approach is weighted score fusion, where each retrieval method is assigned a weight based on performance. For example, semantic retrieval may get a higher weight for contextual queries, while keyword search may be prioritized for exact entity matching. After merging, I would remove duplicates by document ID or semantic similarity threshold. Then, I would optionally apply a reranking stage using a cross-encoder model to refine relevance based on the original user query.



Finally, I would evaluate the merged ranking using metrics like Recall@K, NDCG, and MRR to ensure the fusion strategy improves retrieval quality rather than introducing noise.

Thank you! Follow for more such content.